

MINOR PROJECT REPORT

Entertainment Content Segmentation Using K-Means Clustering

Submitted By

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1. Introduction

With the rapid growth of online streaming platforms such as Netflix, Amazon Prime Video, and Disney+, thousands of movies and television shows are available to users. Organizing this huge amount of content manually is difficult and time-consuming. Machine Learning provides techniques that can automatically discover patterns in data.

Unsupervised learning is particularly useful because it can group similar content without requiring labeled data. In this project, the K-Means clustering algorithm is used to segment entertainment content based on its characteristics such as release year, duration, content type, and rating. These clusters can help improve recommendation systems and assist users in discovering similar content.

2. Problem Statement

Streaming platforms contain a vast collection of entertainment content. It is difficult to manually categorize movies and TV shows based on multiple features. Therefore, an automated clustering technique is required to group similar content for better recommendation and analysis.

3. Objective

The main objectives of this project are:

- To analyze Netflix entertainment content.
 - To preprocess the dataset.
 - To perform Exploratory Data Analysis (EDA).
 - To apply the K-Means clustering algorithm.
 - To evaluate the clustering performance using the Silhouette Score.
 - To identify meaningful groups of entertainment content.
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4. Dataset Description

Dataset Name

Netflix Movies and TV Shows Dataset

Source

Kaggle

<https://www.kaggle.com/datasets/shivamb/netflix-shows>

Dataset Features

Feature	Description
show_id	Unique ID
type	Movie or TV Show
title	Title of content
director	Director
cast	Cast members
country	Country
date_added	Date added to Netflix
release_year	Release year
rating	Content rating
duration	Duration
listed_in	Genre
description	Description

The dataset contains thousands of Netflix movies and TV shows from different countries and genres.

5. Data Preprocessing

Several preprocessing steps were performed before training the model.

Step 1

Loaded the dataset using Pandas.

Step 2

Removed unnecessary columns:

- show_id
- title
- director
- cast
- description

These columns were not useful for clustering.

Step 3

Handled missing values.

Missing values in country, rating, duration, and date_added were filled using suitable values.

Step 4

Converted duration into numeric format.

Example:

90 min → 90

Step 5

Encoded categorical columns.

Label Encoding was used for:

- Type
- Country
- Rating
- Listed_in

Step 6

Feature Scaling

StandardScaler was used to normalize the features before clustering.

6. Exploratory Data Analysis (EDA)

EDA helps understand the structure of the dataset before model training.

The following analyses were performed:

- Dataset shape
- Dataset information
- Missing values
- Histogram of Release Year
- Histogram of Duration
- Cluster Distribution

Observations

- Most Netflix content has been released after 2010.
- Movies are more common than TV shows.
- Most titles have durations between 80 and 120 minutes.
- Ratings such as TV-MA and TV-14 appear frequently.

(Insert screenshots of your histograms here.)

7. Model Used

K-Means Clustering

K-Means is an unsupervised machine learning algorithm used to group similar observations into clusters.

Why K-Means?

- Easy to implement
- Fast
- Efficient for large datasets
- Produces meaningful clusters

The Elbow Method was used to determine the optimal number of clusters. Based on the graph, **5 clusters** were selected.

(Insert Elbow Method graph here.)

8. Training Process

The following features were selected:

- Release Year
- Duration
- Type
- Rating

These features were standardized using StandardScaler.

The K-Means algorithm was trained using five clusters.

After training, every Netflix title was assigned a cluster number.

The clustered data was saved as:

Entertainment_Content_Clusters.csv

9. Evaluation Metrics and Results

The clustering performance was evaluated using the **Silhouette Score**.

The Silhouette Score measures how well each data point fits within its assigned cluster.

A higher score indicates better clustering quality.

Results

- Successfully created **5 clusters**
- Cluster labels assigned to every content item
- Cluster summaries generated
- Cluster visualization plotted

(Insert Cluster Scatter Plot here.)

Interpretation

The model grouped similar entertainment content based on duration, release year, rating, and content type. These clusters can improve recommendation systems by suggesting similar content to users.

10. Conclusion

This project successfully demonstrated the use of **K-Means Clustering** for entertainment content segmentation.

The Netflix dataset was cleaned, preprocessed, and analyzed before applying the clustering algorithm.

The Elbow Method helped determine the number of clusters, while the Silhouette Score evaluated clustering quality.

The resulting clusters can support recommendation systems, personalized content delivery, and business analysis.

11. References

1. Kaggle – Netflix Movies and TV Shows Dataset
 2. Pandas Documentation
 3. Scikit-learn Documentation
 4. Python Documentation
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Theory Questions

Q1. Semantic vs Instance vs Panoptic Segmentation

Semantic Segmentation

Semantic segmentation assigns a class label to every pixel in an image. Objects of the same class are treated as one group.

Example: All cars are labeled as "Car."

Instance Segmentation

Instance segmentation identifies each object separately.

Example: Three cars are labeled as Car 1, Car 2, and Car 3.

Panoptic Segmentation

Panoptic segmentation combines semantic and instance segmentation. It labels both object classes and individual object instances.

Semantic	Instance	Panoptic
Classifies pixels	Identifies each object	Combines both

Q2. What is Fuzzy Logic? How is it different from Boolean Logic?

Fuzzy Logic

Fuzzy Logic allows values between 0 and 1. It represents partial truth.

Example:

Temperature = 0.8 Hot

Boolean Logic

Boolean Logic has only two values:

- True
- False

Difference

Boolean Logic	Fuzzy Logic
Only True or False	Values between 0 and 1
Exact decision	Approximate decision
Used in digital circuits	Used in AI and control systems

Q3. Customer Dataset Question

1. Is this supervised or unsupervised learning?

It is an **unsupervised learning** problem because no labels are provided.

2. Which clustering algorithm would you choose first and why?

I would choose **K-Means Clustering** because it is simple, efficient, and suitable for numerical features such as Age, Annual Income, and Spending Score.

3. What assumptions does K-Means make?

- The number of clusters is predefined.
- Clusters are roughly spherical.
- Clusters have similar sizes.
- Features are numerical.
- Euclidean distance is meaningful for measuring similarity.